QUOC HUY NGUYEN

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Summary

Final-year Computer Science Honors student and Undergraduate Researcher focusing on **3D Vision**, **Spatial Intelligence**, and **Deep Learning**. Experienced in developing geometry-aware 3D object detection models, multi-view calibration systems, and end-to-end computer vision pipelines aimed at perceiving and reasoning about the physical 3D world.

Education

Ho Chi Minh City University of Technology (HCMUT) – VNU-HCM — Ho Chi Minh City, Vietnam
B.Sc. in Computer Science – Honors Program — Expected Graduation: 2027

- **GPA:** 3.8/4.0 (8.88/10.0) · **Honors & Awards:** Academic Excellence Scholarship (3 semesters).
- **Relevant Coursework:** Machine Learning, Deep Learning, Computer Vision, Computer Graphics, Linear Algebra, Probability & Statistics, Data Structures & Algorithms.

Publications

MonoBEN: Monocular 3D Object Detection Beyond Fixed Camera iNtrinsics — Project Page
Quoc Huy Nguyen, Chi Dai Le, Duc Dung Nguyen* — *Accepted at **IEEE RIVF 2026** (Oct 2026)*

- Formulated camera intrinsics as a continuous geometric manifold via a lightweight MLP over log-normalized focal parameters ($r_x = f_x/W, r_y = f_y/H$), hierarchically conditioning visual features and object queries without external LLMs or lookup tables.
- Surpassed paper-reported MonoIA across 9 interpolated focal lengths (**22.77% vs. 22.08% AP_{3D}**, peaking at **24.83%** at 850 px) and reproduced MonoIA† on out-of-range extrapolated focals (**21.01% vs. 18.75%** at 600 px), averaging **21.97% AP_{3D}** over all 17 focals (600–1400 px).
- Improved over base detector MonoCoP in **10 of 12 metrics** on standard KITTI (**37.69% Test Easy AP_{BEV}**, **24.20% Val Moderate AP_{3D}**) while maintaining parameter and FLOP parity (**42.37M parameters, 71.77 GFLOPs**) with zero added overhead.

Research & Industry Experience

AITech Lab (HCMUT) × AISIN Corporation (Tokyo Research Center, Japan) — Jan 2025 – Present
Undergraduate Researcher · Advisors: Dr. Duc Dung Nguyen, Dr. Truong Vinh Truong Duy

- Developed **MonoBEN** (IEEE RIVF 2026), an intrinsic-aware monocular 3D detector with continuous focal manifold encoding.
- Researching **viewpoint-aware and viewpoint-agnostic 3D object detection** across **vehicle**, **roadside/infrastructure**, and **drone** viewpoints, alongside **cross-sensor visual 3D detection** (Cross-Dataset Sensor Alignment).

Anh Khoi Technology — Jun 2026 – Aug 2026
Computer Vision Intern

- Built an **end-to-end multi-camera 3D calibration pipeline** from human-motion videos using **MediaPipe 2D keypoints**, OpenCV, **RANSAC**, and **Ceres Solver bundle adjustment** with automated keyframe selection and confidence weighting.
- Curated a dedicated multi-view test dataset, analyzed calibration error tolerances, and containerized with **Docker**.

Selected Projects

VisionDrive3D – Synthetic Driving-Scene Data & Perception Evaluation — GitHub · Project Page · Dataset
Computer Graphics Course Project

- Built an **OpenGL driving-scene simulator** with configurable traffic flows, urban layouts, and camera viewpoints to synthesize synchronized **RGB, depth, instance segmentation, and 2D/3D bounding box** data with automated COCO/YOLO/KITTI export.
- Benchmarked **YOLO26** and **RT-DETRv2-S** models on synthetic driving scenes across mAP, inference throughput, and latency trade-offs.

Bayesian Wiener-Hunt Deconvolution – Image Super-Resolution & Restoration — GitHub · Interactive Demo
Image Processing & Computer Vision Course Project

- Implemented an **unsupervised Bayesian Wiener-Hunt deconvolution** algorithm in NumPy/SciPy using Fourier-domain Laplacian regularization and an **MCMC (Gibbs Sampler)** with conjugate **Gamma priors** to automatically infer noise and prior precision hyperparameters without manual tuning.

RealEstateText – Transformer-Based Text Classification — Live Demo · Project Page
Deep Learning Course Project

- Fine-tuned **BERT** and **DistilBERT** in PyTorch across 13 hyperparameter configurations, achieving **90.94% accuracy** and **80.88% Macro F1-score**, and deployed an interactive demo on Hugging Face Spaces.

Technical Skills

- **Languages & Frameworks:** Python, C++, SQL, Linux Shell, PyTorch, TensorFlow, NumPy, SciPy, OpenCV.
- **3D Vision & Geometry:** Monocular & multi-view 3D detection, camera intrinsics/extrinsics, 3D calibration.
- **Models & Evaluation:** Transformers (ViT, DETR, BERT), CNNs, YOLO26, RT-DETRv2-S, KITTI (AP_{3D}/AP_{BEV}).
- **Simulation & Tools:** OpenGL, synthetic data (COCO/YOLO/KITTI), MMDetection3D, Ceres, MediaPipe, Docker.